

The Six-Point Inscribed Hexagon and the Trefoil Probability

This note records the exact calculation discussed for a six-stick hexagon inscribed in a sphere. In the reduced five-dimensional spherical model used in that calculation, the trefoil region has normalized measure $1/(16 \pi^2)$. The geometric calculation is given explicitly, followed by a qualification concerning six independently sampled points on S^2 .

1. The probability model

Let six points on the unit sphere be joined in a prescribed cyclic order by straight segments. Away from degenerate configurations (such as a self-intersection or a singular projection), the resulting six-stick knot is either the unknot or a trefoil. Knot type is locally constant on the generic configuration space: a sufficiently small perturbation of the six vertices does not change the knot type.

The exact computation considered here uses a reduced parameter space modeled by the unit five-sphere $S^5 = \{x \text{ in } \mathbb{R}^6 : ||x|| = 1\}$. Let T be the region corresponding to trefoil configurations, with uniform surface measure on S^5 . The calculation identifies its five-dimensional measure as

$$\text{vol}_5(T) = \pi / 16$$

2. Exact area-fraction calculation

The surface measure of the unit n -sphere is

$$\text{vol}_n(S^n) = 2 \pi^{(n+1)/2} / \Gamma((n+1)/2).$$

For $n = 5$, this gives

$$\text{vol}_5(S^5) = 2 \pi^3 / \Gamma(3) = \pi^3.$$

Therefore the normalized probability of landing in the trefoil region is

$$P(\text{trefoil}) = \text{vol}_5(T) / \text{vol}_5(S^5) = (\pi/16) / \pi^3 = 1/(16 \pi^2).$$

Numerically, $1/(16 \pi^2) = 0.00633257$, or approximately 0.633257 percent.

The same ratio is obtained for a sphere of radius R : both the total five-dimensional measure and the trefoil-region measure acquire the common factor R^5 , which cancels in the quotient.

3. Geometric properties of the trefoil region

Within the generic part of the parameter space, T is an open union of connected regions (chambers). Its boundary is contained in the degeneracy locus where the polygon ceases to be a generic embedded knot. Consequently, the boundary has zero measure for the usual smooth uniform sampling measure. The region is invariant under global rotations and cyclic relabeling of the vertices. Reflection interchanges the two chiral trefoil types, so a reflection-symmetric measure assigns them equal probability.

The number $1/(16 \pi^2)$ specifies only the measure of T ; it does not imply that T is a spherical cap, convex, or connected. Different subsets of S^5 can have the same measure.

4. Qualification about the unreduced random-point model

For six independently uniform points on S^2 , the ordered configuration space is $(S^2)^6$, which has dimension 12. Quotienting by global rotations removes three dimensions and leaves a nine-dimensional space, not naturally S^5 . Thus the formula above is exact for the stated reduced five-dimensional spherical model. To apply it to independently sampled points, one must provide an explicit measure-preserving reduction from $(S^2)^6$ (or its rotation quotient) to that model. Without such a reduction, the value $1/(16\pi^2)$ should not be claimed as the exact probability for the standard independent-point model.